2D Finite Element Analysis of a Microstrip Transmission Line

A verified numerical evidence chain for capacitance and impedance

Group C

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MTH321 Numerical PDEs — Project 2

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Problem and engineering goal

What we need

- Compute the quasistatic capacitance per unit length.
- Reuse the same mesh for the all-air capacitance.
- Convert the matched pair into $\varepsilon_{\mathrm{eff}}$ and Z_0.

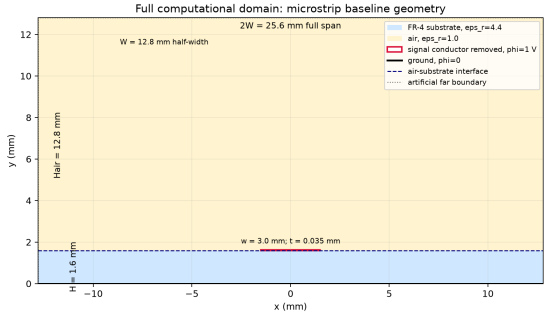
What would not be enough

- A single FEM number near $50\,\Omega$.
- Agreement with an empirical formula only.
- Unchecked corner and boundary effects.

Takeaway

The result matters because the numerical evidence chain makes it defensible.

Geometry and PDE model



Electrostatic model

$$-\nabla \cdot (\varepsilon \nabla \phi) = 0$$

$$\phi = 1\,\mathrm{V} \text{ on trace}, \qquad \phi = 0 \text{ on ground}$$

Homogeneous Neumann conditions are used on the top and side artificial boundaries.

Physical dimensions

$H = 1.6\,\mathrm{mm}$, $w = 3.0\,\mathrm{mm}$, $t = 0.035\,\mathrm{mm}$, substrate $\varepsilon_r = 4.4$.

Weak form and P1 FEM

Weak problem

Find $\phi \in V_g$ such that

$$\int_\Omega \varepsilon \nabla \phi \cdot \nabla v \, d\Omega = 0 \quad \forall v \in V_0.$$

Discretisation

- Conforming triangular P_1 elements.
- Trace interior removed from the field domain.
- Dirichlet dofs eliminated before solving.

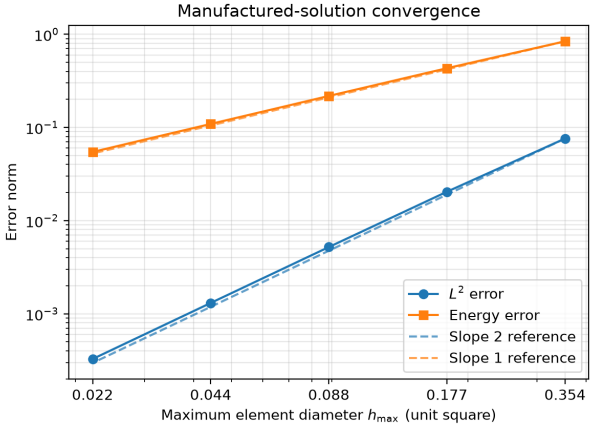
Material interface

ϕ_h is continuous across the air-substrate line. Flux balance is enforced weakly by elementwise integration of discontinuous ε.

Takeaway

The model keeps the conductor boundary, material jump and artificial boundary conditions explicit.

Solver verification: manufactured solution



Known exact solution

$$u(x, y) = \sin(\pi x)\sin(\pi y)$$

Observed rates: $p_{L^2} \approx 2$, $p_E \approx 1$.

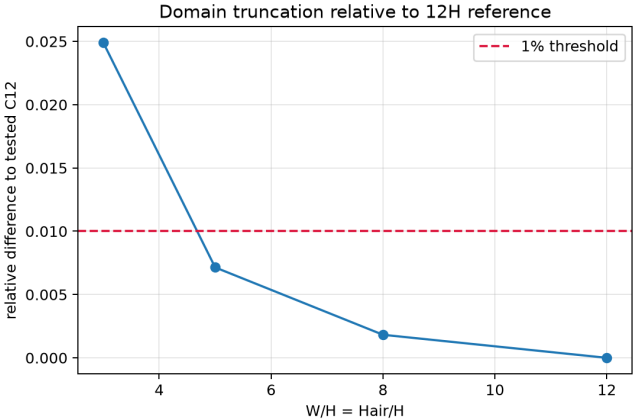
Microstrip baseline checks

Potential	$0 \le \phi_h \le 1$
Symmetry error	3.24×10^{-14}
$\|r_F\|_\infty$	3.10×10^{-25}
δ_{EQ}	2.54×10^{-15}

Takeaway

MMS checks solver accuracy; baseline diagnostics check the assembled microstrip problem.

Domain truncation and the 8H choice



Relative to tested 12H reference

Domain	Difference
3H	2.491%
5H	0.714%
8H	0.182%
12H	reference

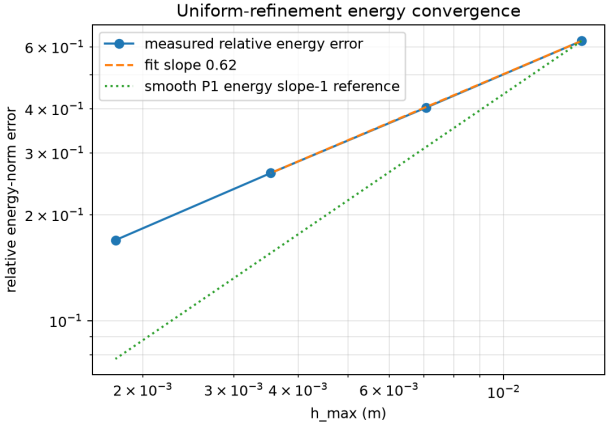
Mesh-adequacy repeat

Finer 12H repeat changed C by about 0.0360%.

Takeaway

The final 8H domain is a controlled cost-accuracy choice, not an assumed infinite-domain result.

Corner singularity and uniform convergence



What limits the rate

- Trace corners concentrate electric field.
- Lower corners also meet the material interface.

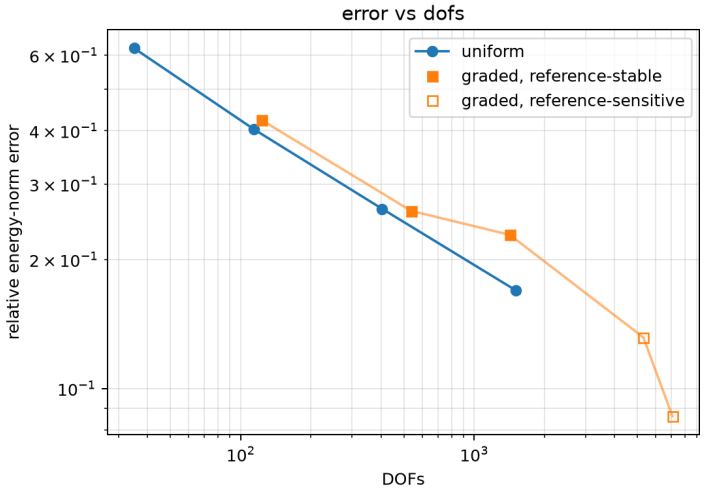
Observed energy fit

$p_E \approx 0.623$

Takeaway

The reduced rate is consistent with corner- and interface-limited regularity.

Uniform vs graded refinement



Cost metrics checked

- DOFs
- NNZ
- Measured median time

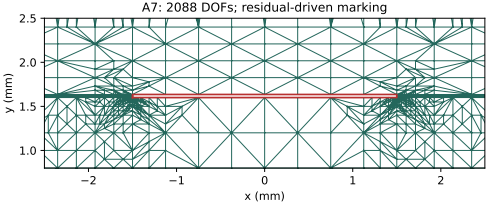
Reference-stable targets

Recorded relative errors: 0.40 and 0.30.

Result

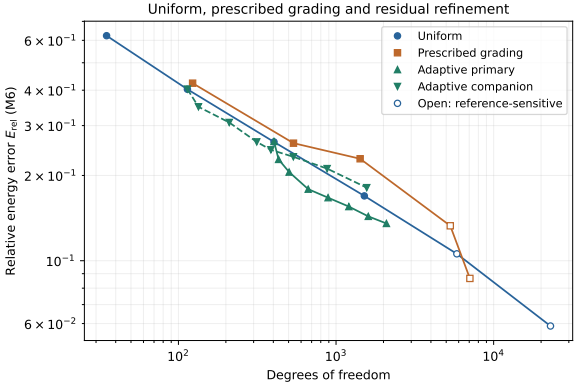
Recorded estimates favoured uniform refinement in DOFs, NNZ and time. A new geometry audit requires common-overlay validation before precise efficiency claims.

Residual-driven adaptive refinement



Solution-driven marking

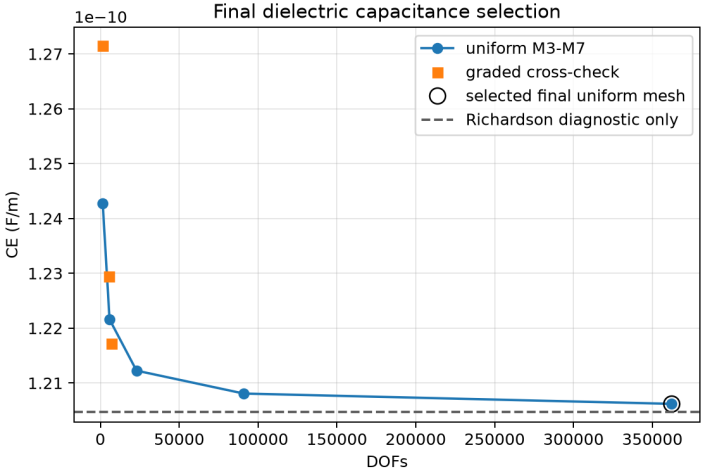
Flux-jump indicator; Dörfler $\theta = 0.5$.
Primary: 404 to 2088 DOFs.
Same $8H$ model, P_1 and direct solver.



Matched-error result: $E_{\mathrm{rel}} = 0.30$

About 18% fewer DOFs than uniform, but 1.83× the cumulative time.
M0-S companion; exact-overlay M6 error with M7 reference check.

Final capacitance selection



Selected result

Uniform M7, 8H domain

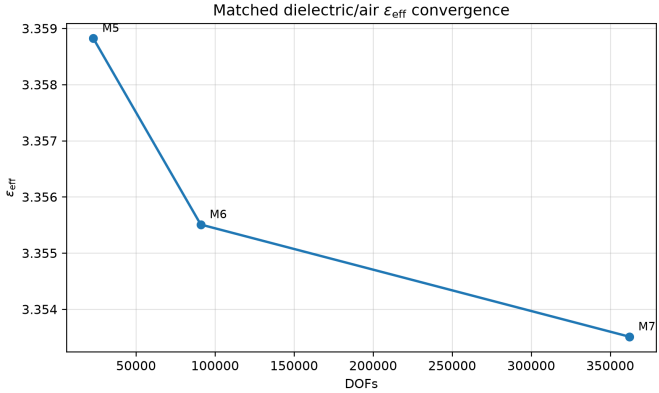
$$C = 120.620\,\mathrm{pF/m}$$

M6→M7 change: 0.156%
Preferred QC target: 0.1%

Why M7?

- Monotone capacitance convergence
- Stable asymptotic trend
- M8 rejected after resource preflight
- Richardson extrapolation is diagnostic only

Matched all-air solve and effective permittivity



Same frozen mesh

The all-air solve reuses the exact M7 mesh from the final dielectric case.

Matched capacitance ratio

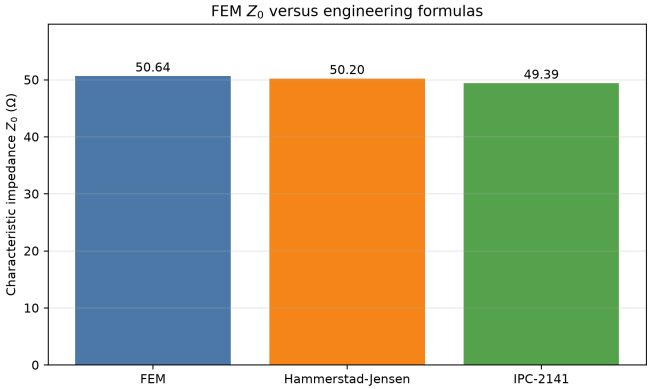
$$C_{\text{air}} = 35.9682\,\text{pF/m}$$

$$\varepsilon_{\text{eff}} = \frac{C}{C_{\text{air}}} = 3.3535$$

Takeaway

Mesh identity protects the ratio C/C_{air} from a hidden remeshing error.

Final Z_0 and engineering benchmarks



Quasi-TEM result

$$Z_0 = \frac{1}{c_0\sqrt{CC_{\mathrm{air}}}} = 50.642\,\Omega$$

HJ-denominator discrepancy

Method	Z_0 (Ω)	$100d_{\mathrm{HJ}}$
FEM	50.642	0.88%
HJ (finite t)	50.201	0
IPC-2141	49.393	1.61%

$$d_{\mathrm{HJ}}(Z) = \frac{|Z - Z_{0,\mathrm{HJ}}|}{|Z_{0,\mathrm{HJ}}|}$$

Takeaway

Engineering formulas support the scale of the answer, while verification and convergence support the FEM result.

Conclusions and limitations

Final reported values

Quantity	Value
C	$120.620\,\mathrm{pF/m}$
C_{air}	$35.9682\,\mathrm{pF/m}$
$\varepsilon_{\mathrm{eff}}$	3.3535
Z_0	$50.642\,\Omega$

What the model leaves out

- Losses, surface roughness and fabrication tolerances are omitted.
- The quasi-static model does not capture full-wave dispersion or radiation.
- Finite-domain and spatial-discretisation sensitivity remain.

Residual adaptivity reduced approximation size at $E_{\mathrm{rel}} = 0.30$, but not cumulative runtime.

Main message

This is not just a FEM result. It is a verified numerical evidence chain from PDE model to engineering impedance.